

$$t = 1.3 \text{ s}$$

4

Ans: C

$$F_n = ma$$

$$U - W = ma$$

With the fuel burning, the mass and the weight of the rocket reduces. Hence, with increase in resultant force and decrease in mass of the rocket, acceleration of the rocket increases.

After 5 s, the mass remains constant. With the resultant force which is only comprised of weight, the acceleration remains constant.

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Ans: D